John Snow LABS

Visual NLP

Alberto Andreotti

October 2026

Agenda

- Introduction to Visual NLP. Basic Concepts. Examples.
- Data Extraction Pipelines. Mixed Formats. Forms. Tables.
- De-identification & Obfuscation. PDFs and Dicoms.
- 5‘ Break
- VLMs. Different sizes and deployment options.
- Handwritten Text.
- Q&A.

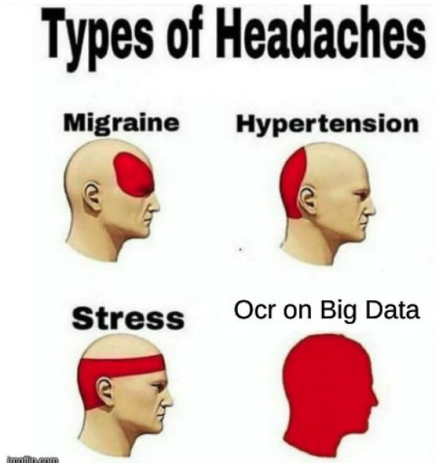
Presenters today



Alberto

Introduction to Visual NLP

- Lots of text data locked into document images.
- We identified the strong need for a scalable solution.
- Three sources of stress: big data, big computation, big models.
- The job is not suitable for a single machine: need to scale out.
- Programming in the cluster is challenging.

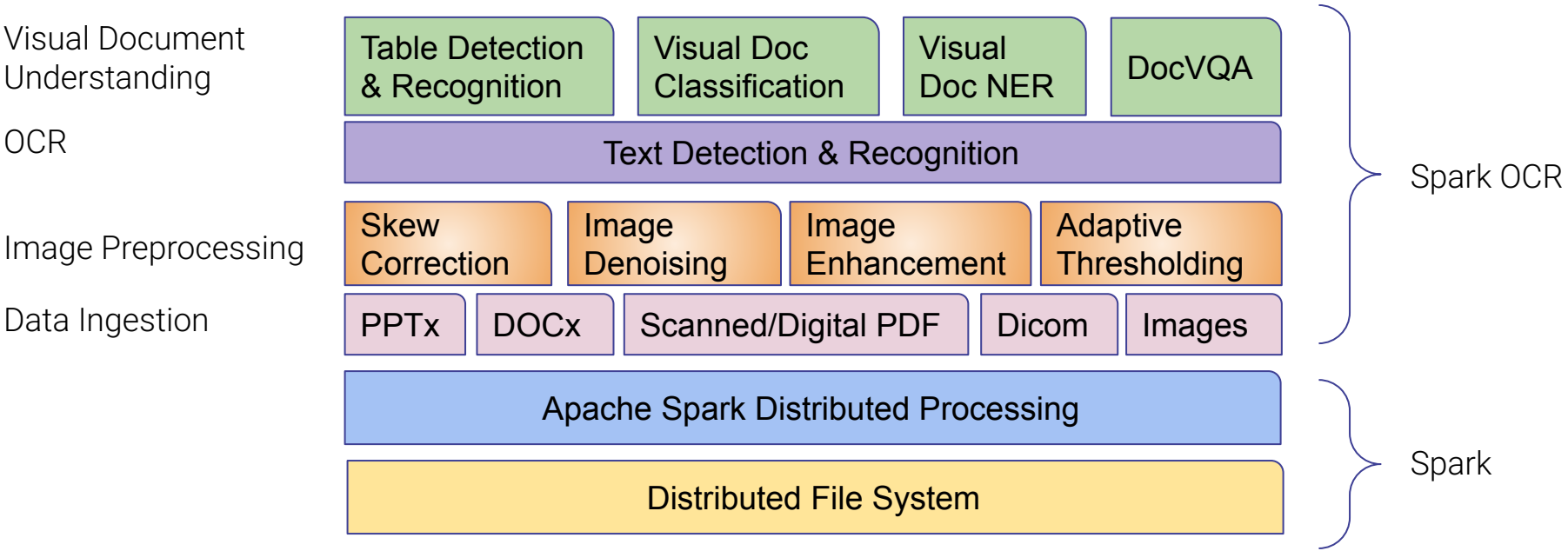


Motivation



STARBUCKS STORE #10208
11302 EUCLID AVENUE
CLEVELAND, OH (216) 229-0749
CHK 664290
12/07/2014 06:43 PM
1912003 DRAWER: 2. REG: 2
VT PEP MOCHA 4.95
SBUX CARD 4.95
XXXXXXXXXXXX3228
SUBTOTAL $4.95
TOTAL $4.95
CHANGE DUE $0.00
---- CHECK CLOSED
12/07/2014 06:43 PM
SBUX CARD X3228 NEW BALANCE: 37.45
CARD IS REGISTERED

- 111835b(JPEG) vs. 315b, a **355** factor!!.
- Density of information is much lower in OCR than NLP.
- Handling images is challenging.


Visual Document Understanding
OCR
Image Preprocessing
Data Ingestion
Table Detection & Recognition
Visual Doc Classification
Visual Doc NER
DocVQA
Text Detection & Recognition
Skew Correction
Image Denoising
Image Enhancement
Adaptive Thresholding
PPTx
DOCx
Scanned/Digital PDF
Dicom
Images
Apache Spark Distributed Processing
Distributed File System
Spark OCR
Spark

Sample Notebooks

+ Basic Layout Example.
+ DocDataloader Pipelines.

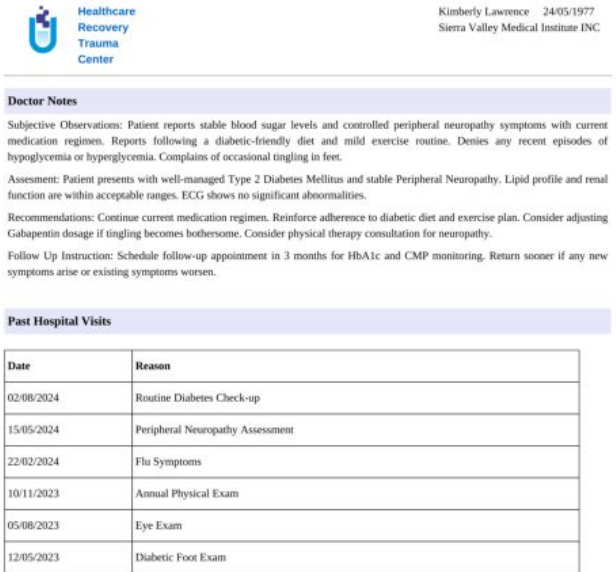
Dataloader Pipelines

Pipeline	Digital PDF	Simple Page	Complex Page	Tables	Forms
doc_data_loader_digital_easy	PdfToText	V4	V4	N/A	N/A
doc_data_loader_digital_hybrid	PdfToText	V1	V4	N/A	N/A
doc_data_loader_digital_hybrid_tables_vlm1	PdfToText	V1	vlm1	vlm1	N/A
doc_data_loader_digital_hybrid_tables_vlm2	PdfToText	V1	vlm2	vlm2	N/A
doc_data_loader_digital_hybrid_tables_forms_vlm1	PdfToText	V1	vlm1	vlm1	vlm1
doc_data_loader_digital_hybrid_tables_forms_vlm2	PdfToText	V1	vlm2	vlm2	vlm2

Source: Visual NLP Release Notes.

De-identification

Field: image_raw
Id: 0
Origin: id_Deidentification_0.pdf
Resolution: 300 dpi
Width: 2480 px
Height: 3507 px
Mode: 5
Number of channels: 3



Healthcare
Recovery
Trauma
Center

Kimberly Lawrence 24/05/1977
Sierra Valley Medical Institute INC

Doctor Notes

Subjective Observations: Patient reports stable blood sugar levels and controlled peripheral neuropathy symptoms with current medication regimen. Reports following a diabetic-friendly diet and mild exercise routine. Denies any recent episodes of hypoglycemia or hyperglycemia. Complains of occasional tingling in feet.

Assesment: Patient presents with well-managed Type 2 Diabetes Mellitus and stable Peripheral Neuropathy. Lipid profile and renal function are within acceptable ranges. ECG shows no significant abnormalities.

Recommendations: Continue current medication regimen. Reinforce adherence to diabetic diet and exercise plan. Consider adjusting Gabapentin dosage if tingling becomes bothersome. Consider physical therapy consultation for neuropathy.

Follow Up Instruction: Schedule follow-up appointment in 3 months for HbA1c and CMP monitoring. Return sooner if any new symptoms arise or existing symptoms worsen.

Past Hospital Visits

Date	Reason
02/08/2024	Routine Diabetes Check-up
15/05/2024	Peripheral Neuropathy Assessment
22/02/2024	Flu Symptoms
10/11/2023	Annual Physical Exam
05/08/2023	Eye Exam
12/05/2023	Diabetic Foot Exam

Field: image_deid
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[redacted]	Peripheral Neuropathy Assessment
[redacted]	Flu Symptoms
[redacted]	Annual Physical Exam
[redacted]	Eye Exam
[redacted]	Diabetic Foot Exam

Deidentification

+ Webinar,
https://github.com/JohnSnowLabs/visual-nlp-workshop/tree/master/webinars/dicom_deid

+ Visual NLP Workshop,
https://github.com/JohnSnowLabs/visual-nlp-workshop/tree/master/jupyter/deid-and-obfuscation

+ Dicom Skill,
https://github.com/JohnSnowLabs/visual-nlp-workshop/blob/master/jupyter/skill/README.md

Examples

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Obfuscation

DeIdentification

SparkOcrDicomDeIdentificationV3

SparkOcrWSIDeidentification

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5’ Break.

VLMs

- Small VLMs inside Visual NLP library. Tables, Forms, Schema Driven Extraction. Handwriting. Deployed through Visual NLP locally.
- Task Specific VLMs. JSL-Vision. Plain & Structured outputs. Deployed locally.
- Medical LLM Small in Databricks.

Local Small VLMs inside Visual NLP

- Cluster-scalable.
- CPU runnable.
- Optimized, quantized.
- Handle a few tasks decently well: OCR, Grounding, Schema constrained extraction, Form Parsing.
- SparkOCRMultiModelVLM.ipynb

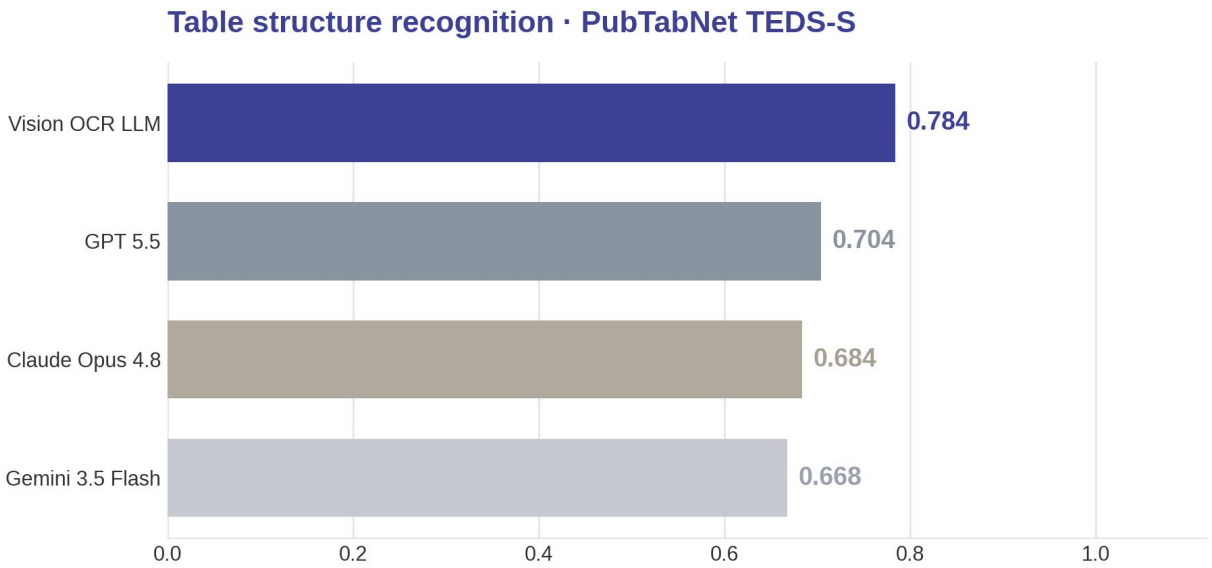
JSL-Vision

- Single Task models, Table Recognition, Bounding Box grounding, Structured Extraction.
- Vision OCR LLM: *Grounding specialist.* Word-level OCR with precise bounding boxes, single-GPU.

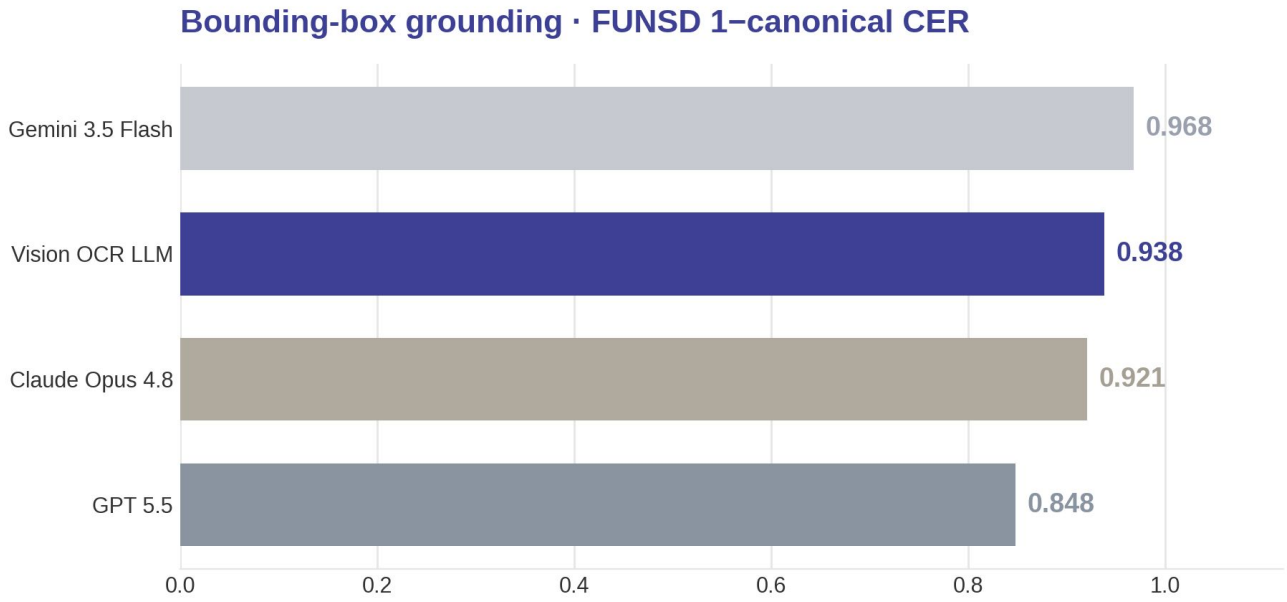
  AWS Marketplace
- Vision OCR Structured LLM: *Structure specialist.* Document to schema-constrained JSON.

  AWS Marketplace
- JSL-Vision Docs.
- Check VisionOCRStructuredLLM.ipynb

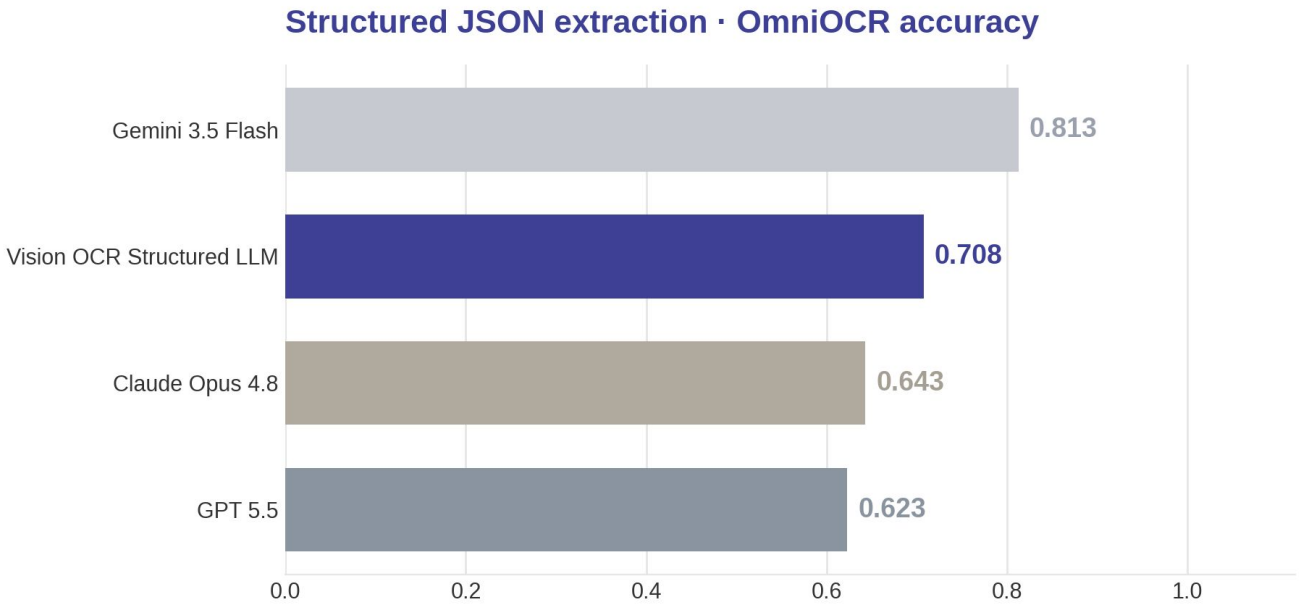
JSL-Vision


Table structure recognition · PubTabNet TEDS-S
Vision OCR LLM
0.784
GPT 5.5
0.704
Claude Opus 4.8
0.684
Gemini 3.5 Flash
0.668
0.0
0.2
0.4
0.6
0.8
1.0

JSL-Vision


Bounding-box grounding · FUNSD 1−canonical CER
Gemini 3.5 Flash
0.968
Vision OCR LLM
0.938
Claude Opus 4.8
0.921
GPT 5.5
0.848
0.0
0.2
0.4
0.6
0.8
1.0

JSL-Vision


Structured JSON extraction · OmniOCR accuracy
Gemini 3.5 Flash
0.813
Vision OCR Structured LLM
0.708
Claude Opus 4.8
0.643
GPT 5.5
0.623
0.0
0.2
0.4
0.6
0.8
1.0

Medical-LLM-Small - Databricks

- Designed to bring medical-grade multimodal intelligence into a smaller, more efficient package.
- medical text but + visual information across scans, charts, diagrams, and structured documents.
- It can analyze X-rays, MRIs, pathology slides, medical diagrams
- 32K context window, compact yet powerful.
- Databricks Marketplace Listing: https://marketplace.databricks.com/details/2d3cabaf-e93e-45e0-a954-822202000afd8/John-Snow-Labs_Medical-LLM-Small
- Notebook

Databricks

```
mlflow.set_registry_uri('databricks-uc')

CATALOG = "john_snow_labs_medical_llm_small_v2"
SCHEMA = "johnsnowlabs_folder"
MODEL_NAME = "jsl-medical-llm-small"

registered_model_name = f"{CATALOG}.{SCHEMA}.{MODEL_NAME}"
```

Sample Notebooks

./basics/SparkOCRDocumentLayoutAnalyzerTextExtraction.ipynb
./basics/SparkOcrDocDataLoaderPipelines.ipynb

./deid/DeIdentification.ipynb
./deid/Obfuscation.ipynb
./deid/SparkOcrDicomDeIdentificationV3.ipynb
./deid/SparkOcrWSIDeidentification.ipynb

./vlms/JSL-Medical-LLM-Small.ipynb
./vlms/VisionOCRStructuredLLM.ipynb
./vlms/SparkOCRMultiModelVLM.ipynb

https://github.com/JohnSnowLabs/visual-nlp-workshop/tree/master/tutorials/october_2026

Contact Us!

Contact us on Slack!

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